Supplementary Materials: DualTCN: A Physics-Constrained Temporal Convolutional Network for Time-Domain Marine CSEM Inversion

Khaled Ahmed[a], Ghada Omar[b]

[a]*School of Computing, Southern Illinois University Carbondale, Carbondale, IL, 62901, USA*
[b]*School of Mathematics and Statistics, Southern Illinois University Carbondale, Carbondale, IL, 62901, USA*

S1. Validation of the irfft Synthesis Pipeline

To verify that the features learned by DualTCN are physically meaningful rather than artefacts of the frequency-to-time conversion, we compare the paper's irfft waveforms against two reference calculations for the five field-representative models (Section 4.5 of the main text) at all four receiver offsets (20 model–receiver pairs total). The first reference corrects the DC misassignment by computing `empymod` responses at 64 frequencies linearly spaced from 0 (using $f = 0.001\,\mathrm{Hz}$ as a near-DC proxy) to $2.0\,\mathrm{Hz}$, isolating the effect of placing the $0.05\,\mathrm{Hz}$ response in the DC bin. The second reference uses a dense 512-frequency grid spanning 0–8 Hz to quantify the combined effect of DC substitution and bandwidth truncation.

The DC misassignment has negligible impact: the Pearson correlation between the paper and corrected-DC normalised waveforms is ≥ 0.9998 for all 20 pairs (mean RMSE $= 0.0005$), and the peak-amplitude ratio lies within 0.988–1.004. This confirms that the galvanic (DC) component is small relative to the inductive response at these offsets and source depths.

The bandwidth effect is larger but still moderate: the paper waveform correlates at $r \geq 0.93$ with the dense reference in 18 of 20 pairs (mean RMSE $= 0.029$), with one outlier at Model D ($\sigma_2 = 0.008\,\mathrm{S\,m^{-1}}$), $r = 100\,\mathrm{m}$, where the highly resistive seafloor shifts significant energy above 2 Hz. Excluding this outlier, the mean correlation is 0.969. A parameter-sensitivity test confirms that a $\pm 5\%$ perturbation in each of the four earth-model parameters produces waveform changes in the same direction in both representations, with the 0.05–2 Hz band retaining 25–100% of the sensitivity of the 0–8 Hz band for all four parameters. The paper's irfft pipeline thus provides a faithful—if bandlimited—representation of the physical signal, and the network's learned features reflect genuine earth-model sensitivity rather than synthesis artefacts. Fig. S1 shows the overlay for all 20 pairs.

S2. Step-Off Source Validation

DualTCN is trained on impulse-response waveforms. Deployment on field data acquired under the step-off convention requires a pre-processing step: division by $\mathrm{i}\,2\pi f$ in the frequency domain (or temporal integration)

Table S1: Step-off source validation: DualTCN predictions on impulse-response inputs (Imp) versus retransformed step-off inputs (RT) for five field-representative models. "Max % diff" is the largest absolute percentage difference between Imp and RT across all four parameters. Units: σ in S/m, d in m.

	Model	$\hat{\sigma}_1^{\mathrm{Imp}}$	$\hat{\sigma}_2^{\mathrm{Imp}}$	$\hat{d}_1^{\mathrm{Imp}}$	$\hat{d}_2^{\mathrm{Imp}}$	$\hat{\sigma}_1^{\mathrm{RT}}$	$\hat{\sigma}_2^{\mathrm{RT}}$	$\hat{d}_1^{\mathrm{RT}}$	$\hat{d}_2^{\mathrm{RT}}$	Max %
A	Shelf	3.04	0.440	80.3	36.2	3.27	0.479	80.8	32.4	10.4
B	Gas hydrate	3.19	0.055	97.1	18.0	3.44	0.081	98.0	17.3	47.2
C	Low contrast	2.51	0.714	58.5	43.2	2.67	0.739	58.6	41.5	6.4
D	Resistive	3.71	0.008	114.2	14.4	3.98	0.017	115.5	12.0	117.3
E	Intermediate	1.73	0.245	90.3	37.6	1.84	0.276	90.9	36.3	12.6

Table S2: DualTCN test-set RMSE converted to physical units (150 000 test samples). e_{norm}: normalised $[0, 1]$ log-space RMSE. $e_{\log}$: RMSE in $\log_{10}$ physical units. $\Gamma = 10^{e_{\log}}$: geometric RMSE factor. "Physical RMSE at geo-mean": $\Gamma - 1$ applied to the geometric mean of the training range.

Parameter	Range	Geo-mean	e_{norm}	$e_{\log}$	Γ	Phys. RMSE
σ_1 (S m^{-1})	0.1–5.0	0.71	0.020	0.034	1.08	0.06 S m^{-1}
σ_2 (S m^{-1})	0.001–1	0.032	0.092	0.276	1.89	0.028 S m^{-1}
d_1 (m)	50–150	87	0.031	0.015	1.04	3 m
d_2 (m)	10–50	22	0.165	0.115	1.30	7 m

before the time series is passed to the network.

To validate this pipeline, we synthesise step-off responses for the five field-representative models and retransform them to the impulse-response representation used by DualTCN. Table S1 compares the resulting predictions ("RT", retransformed from step-off) against predictions obtained directly from impulse-response inputs ("Imp").

For three of the five models (A, C, E), the retransformation introduces differences below 13%, confirming that the pre-processing pipeline is viable under typical shelf and sediment conditions. Model B (gas hydrate) shows a 47% difference concentrated in σ_2 (0.055 vs. 0.081 S/m), where the low seafloor conductivity amplifies relative sensitivity to the retransformation. Model D (resistive basement, $\sigma_2 = 0.008$ S/m) is the worst case: $\hat{\sigma}_2$ shifts from 0.008 to 0.017 S/m (117% relative change), reflecting the fact that highly resistive targets push significant spectral energy above the 2 Hz bandwidth, and the retransformation amplifies the resulting truncation artefact.

S3. Physical-Unit Error Conversion

Table S2 translates the normalised log-space RMSE values into physical units to aid interpretability.

Table S3: Normalised RMSE, total loss, and single-sample latency (A100 GPU, 1 000 runs). Bold = best; † = worse than baseline.

	Variant	Params	σ_1	σ_2	d_1	d_2	Total	ms
	PCRN	638K	0.026	0.110	0.033	0.183	0.1032	2.6
P1	200 ep	638K	0.027	0.102	0.033	0.184	0.0932	2.5
P2	+amp-ratio	638K	0.027	0.101	0.034	0.192	0.0959	2.6
P3†	d_2 wt=5	638K	0.025	0.111	0.035	0.180	0.1145	2.7
P4†	CrossRecv	576K	0.036	0.179	0.036	0.259	0.2469	2.1
P5a	TCN-only	**201K**	0.022	0.109	**0.031**	0.187	0.1020	2.0
P5b†	TF-only	877K	0.097	0.155	0.137	0.224	0.2248	1.4
P5c†	MLP-only	952K	0.091	0.269	0.082	0.272	0.4450	0.5
P6†	Latent=512	770K	0.027	0.105	**0.031**	0.208	0.1007	2.6
P7	MultiScale	384K	0.021	0.100	**0.031**	0.184	0.0931	3.8
P8	TwoStage	401K	0.020	0.098	0.033	0.185	0.0892	3.8
P10a†	iTransf.	381K	0.040	0.180	0.041	0.220	0.2204	1.0
P10b†	PatchTST	369K	0.050	0.202	0.050	0.248	0.2749	1.1
DualTCN		379K	**0.020**	**0.092**	**0.031**	**0.165**	**0.0771**	3.5

S4. Full Ablation Results

Tables S3 and S4 report normalised RMSE, total loss, inference timing, and per-parameter R^2 for all thirteen ablation variants.

S5. Why Attention-Based Encoders Underperform

Both P10a (iTransformer) and P10b (PatchTST) produce total losses above 0.22—more than 2.8× worse than DualTCN and worse than the plain TCN-only baseline.

S5.1. Specific Configurations Tested

We ran exploratory variants of the Transformer-only encoder (P5b) with 1, 2, and 4 layers and 2, 4, and 8 heads (all within ±0.01 total loss of the reported P5b). Beyond two layers, training instability (gradient-norm spikes) was observed. Sinusoidal, learnable absolute, and relative (RoPE-style) positional encodings were compared; none materially affected the final loss. For PatchTST, patch lengths $\{8, 16, 32\}$ were tested; smaller patches marginally improved $R^2_{\sigma_1}$ but not $R^2_{d_2}$.

S5.2. Physical Interpretation

The core difficulty is inductive bias, not hyperparameters. The MCSEM transient is not exchangeable in time: the first 32 samples encode seawater properties through a distinct diffusion regime, while the last 64

Table S4: Per-parameter R^2 and batch-256 throughput. Bold = best.

	Variant	$R^2_{\sigma_1}$	$R^2_{\sigma_2}$	$R^2_{d_1}$	$R^2_{d_2}$	Throughput
	PCRN	0.992	0.855	0.986	0.541	45 K/s
P1	200 ep	0.991	0.876	0.986	0.536	44 K/s
P2	+amp-ratio	0.991	0.877	0.985	0.496	45 K/s
P3	d_2 wt=5	0.992	0.853	0.984	0.560	44 K/s
P4	CrossRecv	0.984	0.615	0.984	0.082	59 K/s
P5a	TCN-only	0.994	0.857	0.987	0.525	127 K/s
P5b	TF-only	0.886	0.713	0.760	0.318	34 K/s
P5c	MLP-only	0.901	0.133	0.915	−0.008	550 K/s
P6	Latent=512	0.991	0.868	0.987	0.411	45 K/s
P7	MultiScale	0.995	0.879	**0.988**	0.539	67 K/s
P8	TwoStage	0.995	0.885	0.986	0.532	67 K/s
P10a	iTransf.	0.980	0.610	0.979	0.342	**247 K/s**
P10b	PatchTST	0.970	0.512	0.968	0.162	69 K/s
DualTCN		**0.995**	**0.898**	0.987	**0.627**	76 K/s

encode the seafloor through exponential decay. Global self-attention treats all positions as equally relevant, which conflates these two physically separate regimes.

The iTransformer collapses each channel to a single token, discarding temporal ordering entirely ($R^2_{d_2} = 0.342$). PatchTST retains locality within each patch but the channel-independent design delays cross-receiver integration. By contrast, DualTCN hardcodes the known temporal structure with a separate encoder branch for the late-time window.

S6. Full Uncertainty Quantification Comparison

Four UQ methods are compared on 5 000 held-out test samples: MC-Dropout ($T = 100$ passes, $p = 0.30$), temperature-scaled MC-Dropout, split conformal prediction ($n_{\text{cal}} = 75{,}000$), and a five-member deep ensemble.

MC-Dropout keeps dropout active at inference and constructs Gaussian intervals from the predictive mean and standard deviation. Temperature scaling learns one scalar τ_k per parameter on the validation set to dilate the MC-Dropout variance. Split conformal prediction uses the empirical residual quantile to construct intervals with provable marginal coverage. The deep ensemble aggregates five independently trained DualTCN models (seeds 0–4).

Table S5 reports the full PICP at six nominal levels, MPIW_{90}, and point-estimate RMSE.

Table S5: PICP at nominal levels $\alpha \in \{50, 70, 80, 90, 95, 99\%\}$ for four UQ methods (5 000 held-out test samples). Bold = closest to nominal for σ_2 and d_2.

α	Method	σ_1	σ_2	d_1	d_2
50%	MC-Dropout	0.688	0.434	0.291	0.280
	Temp-scaled	0.552	0.609	0.488	0.585
	Split conformal	0.504	**0.480**	0.489	**0.496**
	Deep ensemble	0.267	0.214	0.149	0.140
90%	MC-Dropout	**0.944**	0.664	0.596	0.572
	Temp-scaled	0.863	0.845	0.905	0.854
	Split conformal	0.906	**0.900**	**0.902**	**0.905**
	Deep ensemble	0.581	0.480	0.278	0.362
99%	MC-Dropout	0.990	0.785	0.827	0.715
	Temp-scaled	0.956	0.940	0.987	0.947
	Split conformal	**0.993**	**0.989**	**0.990**	**0.991**
	Deep ensemble	0.764	0.656	0.366	0.559
$MPIW_{90}$ (normalised)					
	MC-Dropout	0.078	0.077	0.059	0.135
	Temp-scaled	0.055	0.149	0.112	0.342
	Split conformal	0.062	0.306	0.096	0.500
	Deep ensemble	0.026	0.074	0.018	0.125

S7. Ablation Variant Descriptions

S8. Amplitude-Channel Perturbation Results

S9. Full Conventional Inversion Benchmark

S10. Complete Training Specification

This section provides the full specification of the data generation, input representation, and training protocol to facilitate reproducibility.

S10.1 Forward Model and Geometry

- **Forward operator**: `empymod` v2.3 [31], 1D semi-analytic (digital-filter Hankel transform). Strictly 1D layered earth; no 2D/3D effects.
- **Earth model**: 3 layers — air ($\rho \to \infty$), seawater (σ_1), uniform seafloor half-space (σ_2).
- **Source**: unit-moment horizontal electric dipole (HED) at position $(0, 0, d_1)$ metres below the sea surface, oriented in the x-direction (inline).
- **Receivers**: 4 inline receivers at horizontal offsets $r \in \{20, 50, 100, 200\}$ m from the source, all at depth $z_{\text{obs}} = 20$ m below the sea surface. The receiver component is E_x (inline electric field).

Table S6: Thirteen ablation variants and their modifications relative to the PCRN baseline.

ID	Modification
P1	Doubled training (200 epochs)
P2	Three inter-receiver log-amplitude ratios (11 ch)
P3	d_2 loss weight $\times 5$
P4	Per-receiver TCN + cross-attention fusion
P5a	TCN-only encoder
P5b	Transformer-only encoder
P5c	Flat MLP encoder
P6	Latent dimension doubled to 512
P7	Multi-scale TCN (3 dilation schedules)
P8	Two-stage cascaded prediction
P10a	iTransformer
P10b	PatchTST
DualTCN	Late-time branch + d_{sf} head

- **Frequency sampling**: 64 frequencies linearly spaced from $f_{\min} = 0.05\,\mathrm{Hz}$ to $f_{\max} = 2.0\,\mathrm{Hz}$ ($\Delta f \approx 0.031\,\mathrm{Hz}$). Linear (not logarithmic) spacing.
- **Time-domain conversion**: `numpy.fft.irfft` with output length $N = 128$, yielding $\Delta t = 0.25\,\mathrm{s}$ and a $32\,\mathrm{s}$ time window. No windowing or tapering applied; the $0.05\,\mathrm{Hz}$ response occupies the DC bin (see main text Section 3.1 for discussion).

S10.2 Parameter Sampling

All four parameters are sampled *independently* (no joint correlations) using the following distributions:

- σ_1: log-uniform on $[10^{-1.0}, 10^{0.7}] = [0.10, 5.01]\,\mathrm{S/m}$.
- σ_2: log-uniform on $[10^{-3.0}, 10^{0.0}] = [0.001, 1.0]\,\mathrm{S/m}$.
- d_1: uniform on $[50, 150]\,\mathrm{m}$ (physical space, not log-space).
- d_2: uniform on $[10, 50]\,\mathrm{m}$ (physical space).
- v_0 (source velocity, Doppler correction): uniform on $[0, 100]\,\mathrm{m/s}$.

One million samples are drawn with seed 42 (`numpy.random.default_rng(42)`). The five field-representative models (Section 4.6 of the main text) fall within the training bounds for σ_1, d_1, and d_2; Models B and D have $\sigma_2 = 0.05$ and $0.008\,\mathrm{S/m}$, respectively—both within the training range but in the sparsely sampled low-σ_2 tail.

Table S7: DualTCN R^2 under amplitude-channel perturbations (150 000 test samples). Panel (A): random noise. Panel (B): systematic bias. Bold = clean baseline.

(A) Random noise					
σ_{amp}	$R^2_{\sigma_1}$	$R^2_{\sigma_2}$	$R^2_{d_1}$	$R^2_{d_2}$	$\bar{R}^2$
0.00	**0.995**	**0.898**	**0.987**	**0.627**	**0.877**
0.01	0.919	0.380	0.939	−0.784	0.363
0.02	0.784	−0.112	0.803	−1.465	0.002
0.05	0.368	−1.149	0.280	−2.124	−0.656
0.10	−0.130	−1.941	−0.558	−2.394	−1.256
0.20	−0.683	−2.431	−1.510	−2.671	−1.824
(B) Systematic bias					
β	$R^2_{\sigma_1}$	$R^2_{\sigma_2}$	$R^2_{d_1}$	$R^2_{d_2}$	$\bar{R}^2$
−0.20	0.746	−0.121	0.980	0.494	0.525
−0.10	0.958	0.368	0.990	0.601	0.729
0.00	**0.995**	**0.898**	**0.987**	**0.627**	**0.877**
+0.10	0.951	0.626	0.985	0.273	0.709
+0.20	0.822	0.351	0.974	−0.079	0.517

S10.3 Input Representation and Normalisation

For each receiver $j \in \{1, \ldots, 4\}$:

1. Compute the complex frequency-domain response $E_j(f_k)$ via empymod for $k = 1, \ldots, 64$.
2. Apply irfft to obtain $g_j(t_n)$ for $n = 0, \ldots, 127$.
3. Record the peak amplitude: $A_j = \max_n |g_j(t_n)|$.
4. Peak-normalise the waveform: $\tilde{g}_j(t_n) = g_j(t_n)/A_j$.
5. Compute $\ell_j = \log_{10}(A_j)$.
6. Construct two input channels: channel $2j-1 = \tilde{g}_j(t_n)$ (128 samples), channel $2j = \ell_j$ broadcast to 128 samples.

The resulting input tensor has shape $(8, 128)$. Target parameters are normalised to $[0, 1]$ in $\log_{10}$ space using the bounds in Table 1 of the main text.

S10.4 Noise Injection Protocol

During training only (validation and test sets are clean unless otherwise noted):

- **Waveform noise (all variants)**: additive Gaussian noise $\epsilon_n \sim \mathcal{N}(0, \sigma_w^2)$ added to the normalised waveform channels only. The standard deviation σ_w is drawn log-uniformly from $[10^{-3}, 10^{-1}]$ per sample (corresponding to ∼11–51 dB SNR). The log-amplitude channels are *not* perturbed.

Table S8: DualTCN versus conventional inversion (same `empymod` forward operator). All multi-start runs use 8 starts. Bold = best in each column.

Method	$R^2_{\sigma_1}$	$R^2_{\sigma_2}$	$R^2_{d_1}$	$R^2_{d_2}$	$\bar{R}^2$	s/sample	Evals
Untuned baselines (500 samples)							
NLS-LM (1 start)	0.592	−0.292	0.378	−0.553	0.031	2.75	49
NLS-LBFGSB (1 start)	0.465	−1.077	0.337	−0.927	−0.301	8.01	291
NLS-LM (8 starts)	0.938	−0.330	0.961	−1.053	0.129	12.09	338
NLS-LBFGSB (8 starts)	0.961	0.179	0.980	−0.362	0.439	73.58	2,630
Occam regularised (200 samples)							
Occam-LM (λ=0.01)	0.730	−0.385	0.853	−1.030	0.042	14.71	170
Improved baselines (500 samples)							
TT-LM	0.938	−0.330	0.961	−1.053	0.129	13.00	378
TIK-LBFGSB	0.962	0.263	0.982	−0.296	0.478	123.14	4,417
WS-LM (DualTCN init)	0.993	0.689	0.997	0.343	0.756	13.22	380
WS-LBFGSB (DualTCN init)	0.993	0.781	**0.998**	0.364	0.784	92.80	3,312
Neural network							
DualTCN	**0.996**	**0.884**	0.988	**0.578**	**0.862**	**0.0035**	1

- **Amplitude noise (DualTCN-AmpAug only)**: curriculum training. Epochs 1–20: clean. Epochs 20–40: linear ramp of σ_{amp} from 0 to the target range. Epochs 40–100: $\sigma_{\text{amp}} \sim U[0.001, 0.01]$ in $\log_{10}$ units applied independently per receiver.
- **Colored noise (DualTCN-Colored only)**: $1/f$ (pink) noise spectrum replacing white Gaussian on the waveform channels, generated via spectral shaping.
- **Per-receiver bias (DualTCN-RecvBias only)**: additive bias $\beta_j \sim U[-0.03, +0.03]$ $\log_{10}$ applied independently per receiver to the log-amplitude channels.

S10.5 Training Hyperparameters

- **Optimiser**: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\lambda = 10^{-3}$, weight decay 10^{-3}).
- **Learning rate schedule**: one-cycle cosine (5-epoch linear warmup, peak rate 5×10^{-4}, final rate 5×10^{-6}; final-div-factor = 100).
- **Batch size**: 256.
- **Epochs**: 100 (no explicit early stopping; best validation-loss checkpoint selected).
- **Dataset split**: 700 K train / 150 K val / 150 K test (70/15/15, sequential split).
- **Loss**: $\mathcal{L}^{\text{prof}}_{\text{MSE}} + 2\sum_{i=1}^{4} w_i \mathcal{L}^{(i)}_{\text{Huber}}$; Huber $\delta = 0.1$; weights $\mathbf{w} = [1, 3, 3, 2]$ for $[\sigma_1, \sigma_2, d_1, d_2]$.

- **Gradient clipping**: max norm = 1.0.
- **Hardware**: single NVIDIA A100-SXM4-40 GB GPU.
- **Training time**: ∼3–4 GPU-hours per variant.
- **Framework**: PyTorch 2.x; `torch.compile` and automatic mixed precision (AMP) enabled.
- **DualTCN auxiliary loss**: Huber($\hat{d}_{\mathrm{sf}}$, $d_{\mathrm{sf}}^{\mathrm{true}}$) with weight 0.5, added to the main loss. d_{sf} normalised as $(\log_{10}(d_{\mathrm{sf}}) - 1.778)/0.523$.

S10.6 Domain of Applicability

DualTCN's predictions are valid under the following conditions:

1. The subsurface is well approximated by a laterally uniform 1D layered model beneath the receiver spread (< 200 m lateral extent). Lateral heterogeneity not represented in training.
2. Earth-model parameters fall within the training ranges (Table 1). Extrapolation beyond these ranges is untested.
3. The acquisition geometry matches the training configuration (four inline receivers at 20–200 m offsets, 20 m depth). Different geometries require retraining on matched synthetics.
4. The source waveform is either impulse (direct compatibility) or step-off (requires $1/(\mathrm{i}2\pi f)$ pre-processing; validated for $\sigma_2 \geq 0.05$ S/m, degraded for resistive targets).
5. Amplitude calibration uncertainty is $\leq$2% per receiver (for the unaugmented variant) or $\leq$5% (for DualTCN-AmpAug).

S11. Extended Background and Related Work

This section provides the full background and related work discussion that was condensed in the main text.

S11.1 Sensitivity Structure of MCSEM Data

How much information an MCSEM measurement carries about the subsurface depends on both the source–receiver offset and the time window being examined. Receivers close to the source are dominated by the direct arrival and the airwave, so they tell us relatively little about what lies beneath the seafloor. Receivers at larger offsets, by contrast, record the guided wave that has interacted with the resistive sub-seafloor, and they can therefore resolve conductivity contrasts at depth [16].

In the time domain, this offset dependence translates into a temporal hierarchy. Early-time samples are sensitive mainly to the water column (its conductivity σ_1 and depth d_1), while late-time samples—where the

signal decays approximately as $\exp(-2u_1 d_2)$, with u_1 the vertical wavenumber in seawater—carry information about the seafloor (σ_2 and the source-to-seafloor distance d_2). Recording the full transient preserves information across a continuous range of diffusion times, in contrast to frequency-domain measurements, which sample only a discrete set of frequencies [9].

This temporal hierarchy is central to the architectural decisions we make in this work. If different time windows contain information about different parameters, then a network that treats all time samples identically is leaving information on the table; a better design would give the network the ability to specialise.

S11.2 Classical 1D MCSEM Inversion and Acquisition Variants

The gold standard for 1D MCSEM inversion is the Occam algorithm [11], which minimises a data misfit subject to a smoothness constraint on the resistivity–depth profile, producing the simplest model consistent with the data. One-dimensional Occam inversion of multi-component, multi-frequency CSEM data was formalised by Key [15], who provided analytic expressions for the Jacobians of the 1D layered forward operator that make the inversion computationally efficient and accurate. The two-layer parameterisation used in this study (seawater conductivity σ_1, seafloor conductivity σ_2, source depth d_1, gap d_2) is precisely the minimal model that Occam reduces to when applied to shallow-water, single-interface geology; it is also the standard starting model for more complex iterative schemes [10, 16].

A practical limitation of Occam and related regularised methods is computational cost: even with analytic Jacobians each sample requires dozens to hundreds of forward evaluations, making real-time interpretation during acquisition infeasible. Constable and Srnka [10] and Key [16] discuss these operational constraints and the consequent demand for faster inversion strategies.

The majority of published MCSEM surveys use a *seabed-deployed* geometry: receivers are placed on the ocean floor and the source is towed just above them. A geometrically distinct variant, *towed-streamer EM* (TSEM), tows both source and receivers at the sea surface [34], trading the deep-seafloor coupling of the seabed geometry for continuous coverage and rapid turnaround.

DualTCN is trained and evaluated using a *shallow-tow* configuration in which the receivers are modelled at a fixed depth of $z_{\mathrm{obs}} = 20\,\mathrm{m}$ below the sea surface—i.e., within the water column, not on the seafloor. This is an intermediate geometry between seabed-deployed and surface-towed: the source operates at 50–150 m depth (within or near the seafloor interface) while the receivers are above it, at horizontal offsets of 20–200 m. The electromagnetic field of a horizontal electric dipole (HED) moving at constant speed within such a layered conducting medium—a horizontal layer bounded by two homogeneous half-spaces representing seawater and seafloor—was analysed by Sami [28], whose formulation provides the physical basis for the source model used here. Compared with seabed receivers ($z_{\mathrm{obs}} \approx d_{\mathrm{sf}}$), water-column receivers at 20 m depth are further from the seafloor interface and closer to the air–sea boundary, which reduces the amplitude of the seafloor-refracted signal and increases sensitivity to the airwave, particularly at near offsets. This geometry

choice was motivated by the computational simplicity of a single, fixed observation depth, but the sensitivity differences relative to a seabed-deployed survey should be borne in mind when interpreting the reported accuracies. The neural network framework itself generalises to any source–receiver geometry via retraining on geometry-matched synthetic data; applying DualTCN to a seabed-deployed or TSEM configuration would require regenerating the training set with the appropriate z_{obs} and offset ranges.

S11.3 Deep Learning for Electromagnetic Inversion

Deep learning has been applied to electromagnetic inversion with increasing scope. Puzyrev [25] demonstrated CNN-based 1D frequency-domain CSEM inversion; Puzyrev and Swidinsky [26] extended this to joint frequency- and time-domain frameworks. Araya-Polo et al. [3] showed that neural-network initial models reduce conventional solver iterations from fifty to single digits, and Liu et al. [19] achieved competitive accuracy for resistivity inversion at a fraction of the cost. Zhang et al. [33] trained 3D convolutional models on $>$ 500,000 CSEM samples.

In airborne time-domain EM (ATEM), physics-guided strategies closely related to DualTCN have emerged: differentiable physics decoders, disentangled multi-branch encoders, and auxiliary depth losses [21, 20, 8]. DualTCN adapts these ideas to the marine CSEM setting (Section S16). Dilated TCNs [6] provide the temporal inductive bias, and physics-informed neural networks [27] offer a complementary approach.

Beyond supervised networks, several alternative paradigms have emerged. Projected Gauss–Newton methods with learned priors [1] integrate data-driven regularisation into iterative solvers, retaining explicit data-misfit control at the cost of per-sample iteration. Test-time implicit-prior methods such as Deep Image Prior (DIP) [30] have been adapted for geophysical inversion, producing regularised estimates without labelled training data; Sun and Li [29] demonstrated this for magnetotelluric (MT) inversion, and unsupervised differentiable forward-operator approaches have been explored for MT by Bai et al. [5]. Ensemble Kalman inversion [14] provides a derivative-free alternative for low-dimensional inverse problems, naturally producing uncertainty estimates through ensemble spread; it is particularly well suited to compact 1D parameterisations similar to ours but scales poorly to the million-sample throughput required for real-time survey processing. The OpenEM initiative [23] provides public EM datasets and benchmark models, though its current scope emphasises onshore and airborne regimes rather than the marine time-domain setting targeted here. For uncertainty quantification, invertible neural networks [4], normalising flows [24, 7], and deep ensembles [17] represent the current state of the art for posterior estimation in geophysical inversion.

S11.4 Open Questions

Six significant gaps remain in the literature despite the progress described above.

The first is that every deep-learning marine CSEM inversion published to date works in the frequency domain [25, 26, 33, 32, 18]. Time-domain MCSEM data, which record causal transient responses and are

increasingly used in shallow-water and pulsed-source configurations, have not been addressed. (Puzyrev and Swidinsky [26] includes a time-domain component, but it targets land-based TEM, not marine CSEM.) The causal, multi-channel structure of these transients is well suited to temporal convolutional architectures, but this connection has not been exploited.

The second gap concerns the output representation. Existing methods uniformly predict a discretised resistivity profile consisting of 50 to 200 independent depth samples, an output space that admits non-monotonic and geologically implausible profiles. For the two-layer earth model we consider here, just four scalars fully specify the subsurface. No prior work has regressed these parameters directly or enforced the resulting profile structure through a differentiable analytic decoder.

Third, no systematic comparison of encoder architectures—dilated TCN, global Transformer, flat multilayer perceptron (MLP)—has been carried out for multi-receiver time-domain MCSEM inversion. Most studies adopt a single architecture without empirical justification.

Fourth, the source-to-seafloor distance d_2 is encoded mainly in the late-time diffusion tail. No existing architecture dedicates a separate encoder branch to this regime, and no method has introduced auxiliary physical objectives (such as the total seafloor depth $d_{\mathrm{sf}} = d_1 + d_2$) to improve the gradient signal for d_2.

Fifth, although physics-guided deep-learning strategies developed in the ATEM community (disentangled multi-window encoders, auxiliary depth losses, differentiable physics decoders) are directly relevant to marine MCSEM, no study has transferred or evaluated these ideas in the marine setting.

Sixth, all existing deep-learning MCSEM inversion methods produce point estimates without calibrated uncertainty. Probabilistic methods capable of characterising the full posterior—invertible neural networks, conditional normalising flows, and deep ensembles—have been demonstrated for seismic and potential-field inversion but have not been applied to marine MCSEM.

DualTCN addresses gaps one through four directly, draws on ATEM methodology to address gap five, and provides a first MC-Dropout uncertainty baseline—together with post-hoc calibration and conformal prediction comparisons—that motivates the probabilistic extensions identified in gap six.

S12. Extended Earth Model and Forward Modeling Details

This section provides the detailed forward modeling content that was condensed in the main text (Section 3.1).

S12.1 Frequency Band

The 0.05–2.0 Hz band is chosen to span the diffusion depth range relevant to our geometry and parameter space. The electromagnetic skin depth in a medium of conductivity σ at frequency f is $\delta = \sqrt{2/(\omega\mu_0\sigma)}$. For seawater ($\sigma_1 \approx 3\,\mathrm{S\,m^{-1}}$) at $f_{\min} = 0.05\,\mathrm{Hz}$, $\delta \approx 1{,}300\,\mathrm{m}$—well beyond the seafloor depths considered

($d_{\mathrm{sf}} \leq 200\,\mathrm{m}$), ensuring the lowest frequency captures the full late-time diffusion signature that carries information about σ_2 and d_2. At $f_{\max} = 2.0\,\mathrm{Hz}$ in seawater, $\delta \approx 205\,\mathrm{m}$, adequate to resolve the early-time interactions with the shallowest sources considered ($d_1 \geq 50\,\mathrm{m}$). Frequencies above 2 Hz are dominated by the direct arrival and the geometric-spreading regime rather than the diffusion tail, and carry diminishing information about the subsurface at offsets of 20–200 m; the band is therefore both physically motivated and computationally efficient.

S12.2 DC Component and High-Frequency Truncation

The lowest computed frequency is $f_{\min} = 0.05\,\mathrm{Hz}$, not DC ($f = 0$). When passed to `irfft`, the bin at index $k = 0$ is interpreted as the DC response; it instead holds the 0.05 Hz response. The true galvanic (DC) field, which arises from charge accumulation at conductivity contrasts, is therefore absent from the synthesised time series. At the source–receiver offsets used here (20–200 m) and source depths ($d_1 = 50$–$150\,\mathrm{m}$), the DC field is small relative to the inductive component across our conductivity range, so this approximation has a minor effect on the field amplitude at $t \ll 1/f_{\min} = 20\,\mathrm{s}$. The spectral truncation at 2 Hz acts as an ideal low-pass filter, suppressing content above the Nyquist frequency. The corresponding rectangular spectral window creates Gibbs-like ringing at early times ($t \lesssim \Delta t = 0.25\,\mathrm{s}$); because both the training signals and any evaluation signal are constructed identically, the network adapts to this representation and the ringing is not a source of bias.

S12.3 Wrap-Around and Aliasing

The `irfft` output is a discrete, periodic sequence with period $T = N\,\Delta t = 32\,\mathrm{s}$. If the transient has not decayed to negligible amplitude by $t = 32\,\mathrm{s}$, the periodic extension causes temporal aliasing. The late-time MCSEM transient decays as $E(t) \sim t^{-5/2}$ for a dipole source in a uniform half-space; at the receiver offsets considered (20–200 m), the signal amplitude at $t = 32\,\mathrm{s}$ is of order $(32\,\mathrm{s}/t_{\mathrm{peak}})^{-5/2}$ times the peak value, where t_{peak} is at most a few seconds. This gives a fractional wrap-around amplitude below 10^{-3} for all parameter combinations in our training set, confirming that aliasing is negligible.

S12.4 Implicit Source Waveform

`empymod` computes $E(\mathbf{r}, f)$, the complex frequency-domain electric field for a unit-moment harmonic dipole source. Applying `irfft` to this spectrum (without dividing by $\mathrm{i}\,2\pi f$) produces the time-domain impulse response $g(t)$—the field generated by a broadband Dirac-delta current pulse. This differs from the step-off convention widely used in time-domain MCSEM practice [9], for which the transient is $E_{\mathrm{step}}(t) = \int_t^\infty g(\tau)\,\mathrm{d}\tau = \mathcal{F}^{-1}\big[E(f)/(i2\pi f)\big]$. The impulse-response convention is self-consistent across training and evaluation—every sample is synthesised and inverted using the same procedure—so the network correctly learns to invert this representation. Deployment on field data acquired under a step-off or square-wave source

protocol would require a pre-processing step (temporal integration of the received transient, or equivalently division by $i2\pi f$ in the frequency domain) before the time series is passed to DualTCN; Section S2 validates this pipeline on five field-representative models.

S12.5 Synthesis Pipeline: Known Simplifications

Three deliberate simplifications distinguish the synthesis pipeline from a production MCSEM processing flow and should be borne in mind when interpreting the reported accuracies. *First*, the DC component is absent: the lowest computed frequency is $f_{\min} = 0.05\,\mathrm{Hz}$, so the galvanic response is not modelled. The supplementary validation (Section S1) confirms the effect is negligible ($r \geq 0.9998$ versus a corrected-DC reference). *Second*, the 2 Hz bandwidth truncation acts as a low-pass filter; a dense 8 Hz reference shows moderate differences (mean $r = 0.97$), with the largest impact on highly resistive targets ($\sigma_2 < 0.01\,\mathrm{S\,m^{-1}}$) where significant energy lies above 2 Hz. A production system targeting resistive reservoirs should extend the band to 4–8 Hz. *Third*, the receiver geometry ($z_{\mathrm{obs}} = 20\,\mathrm{m}$, offsets 20–200 m) differs from standard seabed-deployed surveys; generalising to other geometries requires retraining on geometry-matched synthetics. Despite these simplifications, the pipeline is self-consistent—every sample is synthesised and inverted identically—so the network learns to invert this representation faithfully, and the accuracy metrics reflect genuine sensitivity to earth-model parameters rather than synthesis artefacts.

S12.6 Two-Channel Input Design

Each receiver trace is normalised by its peak amplitude to remove the offset-dependent amplitude scaling that varies between surveys due to differences in source strength, coupling, and geometric spreading. The logarithm of the peak amplitude (computed from the clean signal before noise injection) is appended as a second channel, preserving the absolute signal level in a form that is better conditioned for gradient-based optimisation. This two-channel design is deliberate: the normalised waveform shape carries information about the relative timing and decay of the transient (primarily sensitive to σ_1, d_1, and the ratio σ_1/σ_2), while the log-amplitude channel encodes the absolute signal strength, which is the primary discriminating feature for σ_2 and d_2. Separating these two information streams improves noise robustness for the *waveform* channel: in field acquisition the peak amplitude is estimated from the stacked average of many repeated transmissions, so the log-amplitude channel experiences substantially lower effective noise than any individual waveform realisation.

However, this design creates a known vulnerability: because σ_2 and d_2 depend almost entirely on the log-peak-amplitude channel (one scalar per receiver after stacking), any uncompensated error in that scalar—from residual stacking noise, source-strength variation, receiver coupling changes, or calibration drift—directly corrupts the two geophysically most important outputs. The amplitude noise experiments (Section S8 of this Supplement) quantify the severity of this vulnerability; it is a direct consequence of the representation choice,

Table S9: DualTCN predictions for five field-representative 1D earth models. Forward responses were synthesised with `empymod` using the training acquisition geometry; DualTCN was applied in a single forward pass with no fine-tuning or parameter adjustment. Mean absolute percentage error across all parameters and models: 12.1%. Units: σ in S/m, d in m. Superscript * = true (reference) value.

	Model	σ_1^*	$\hat{\sigma}_1$	σ_2^*	$\hat{\sigma}_2$	d_1^*	$\hat{d}_1$	d_2^*	$\hat{d}_2$
A	Shelf	3.20	3.11	0.500	0.617	80	80.5	30	29.6
B	Gas hydrate	3.00	3.31	0.050	0.037	100	96.5	20	18.0
C	Low contrast	2.50	2.69	0.700	0.638	60	56.7	40	45.2
D	Resistive basement	3.50	3.85	0.008	0.005	120	115.1	15	11.7
E	Intermediate	2.00	1.78	0.250	0.258	90	90.6	25	36.6

not an incidental artefact. The resulting input tensor has dimensions $(8, 128)$: eight channels (four receivers $\times$ two channels) by 128 time samples.

S13. Field Validation and Survey Benchmarks

S13.1 Validation on Field-Representative Models

To assess DualTCN on earth models representative of field conditions, we selected five canonical 1D models whose parameter combinations span the range reported in published shallow-water MCSEM studies [9, 16]: a typical shelf background, a gas-hydrate resistive layer, a low-contrast sediment, a resistive basement, and an intermediate mixed-sediment scenario (Table S9). Forward responses were computed with `empymod` using the identical acquisition geometry as the training data (receivers at 20, 50, 100, 200 m; $z_{\text{obs}} = 20\,\text{m}$), ensuring a controlled comparison free of pre-processing artefacts. DualTCN was applied in a single forward pass with no retraining or fine-tuning.

Table S9 reports the predicted parameters and their absolute errors. σ_1 is recovered to within 0.09–0.35 S/m (3–11%) and d_1 to within 0.5–4.9 m (1–6%) across all five models, consistent with the test-set performance. The seafloor conductivity σ_2 is harder: the largest relative error (33%) occurs for model D, where the high resistivity ($\sigma_2 = 0.008\,\text{S/m}$) pushes the signal toward the noise floor of the late-time trace. The source-to-seafloor distance d_2 shows the greatest spread: model A (typical shelf) is recovered to within 0.4 m (1%), while model E (intermediate, $\sigma_1/\sigma_2 = 8$) produces the largest absolute error of 11.6 m (46%), consistent with the network's tendency to underestimate d_2 when the conductivity contrast is moderate and the late-time signal energy is limited. Fig. S2 shows the true and predicted conductivity profiles; the two-step shape is recovered in every case, with the main depth discrepancy concentrated in model E.

S13.2 Step-Off Source Validation

Most time-domain MCSEM surveys use a step-off or square-wave source protocol rather than the impulse convention used by DualTCN. Deployment requires a pre-processing step: division by $\mathrm{i}\,2\pi f$ in the frequency domain converts step-off data to the impulse-response format. We validate this pipeline on the five field-representative models (Section S13.1), synthesising step-off responses, retransforming to impulse form, and comparing predictions against those obtained from direct impulse-response inputs (Table S3).

The retransformation introduces $< 13\%$ prediction differences for moderate-to-high seafloor conductivities ($\sigma_2 \gtrsim 0.05\,\mathrm{S/m}$) but up to 117% for the highly resistive model D ($\sigma_2 = 0.008\,\mathrm{S/m}$), where bandwidth truncation at 2 Hz amplifies the conversion artefact.

Mitigation strategies for resistive scenarios.. Three approaches can address the bandwidth limitation for resistive targets, listed in order of increasing modification: (i) *Broader bandwidth synthesis*: extending the frequency band to 4–8 Hz would capture the high-frequency content that carries σ_2 information for resistive seafloors. The $1/(\mathrm{i}2\pi f)$ division that converts step-off to impulse form amplifies low-frequency noise; applying a minimum-frequency floor ($f_{\min} = 0.01\,\mathrm{Hz}$) or Tikhonov regularisation ($|E(f)|^2 + \epsilon$ in the denominator) stabilises the deconvolution at negligible accuracy cost for $f > 0.05\,\mathrm{Hz}$. (ii) *Constrained deconvolution with physical priors*: imposing monotonic decay and positivity constraints on the impulse response during the f-domain conversion can suppress the ringing artefacts that arise from spectral truncation. (iii) *Direct training on step-off waveforms*: bypassing the impulse conversion entirely by training DualTCN on $E_{\mathrm{step}}(t)$ removes the $1/(\mathrm{i}2\pi f)$ amplification altogether; this requires regenerating the training dataset but no architectural changes. Approach (iii) is the most robust and is our recommended path for production deployment. The current impulse-response convention was chosen for consistency with the ATEM literature and to facilitate direct comparison with published architectures.

S13.3 Benchmark Against Published Survey Models

To assess DualTCN on geologically realistic scenarios beyond the generic five-model validation (Section S13.1), we benchmark against seven 1D earth models derived from published shallow-water MCSEM surveys at sites spanning the Gulf of Mexico, offshore Australia, the North Sea, the Norwegian margin, West Africa, and offshore Brazil [10, 15, 9, 16]. Forward responses are synthesised with `empymod` using the training acquisition geometry; DualTCN is applied in a single forward pass with no retraining.

Table S10 reports the results. For models with moderate-to-high seafloor conductivity ($\sigma_2 \geq 0.3\,\mathrm{S/m}$), the mean absolute percentage error (MAPE) is 7.5%—consistent with the test-set accuracy. For highly resistive targets ($\sigma_2 < 0.05\,\mathrm{S/m}$), the unaugmented DualTCN produces large σ_2 errors (up to 1 125% for the Brazil pre-salt analog at $\sigma_2 = 0.005\,\mathrm{S/m}$), confirming the failure-mode analysis.

To address this, we train DualTCN-Weighted: a variant with inverse-σ_2 sample weighting ($w_i = 1/(\sigma_{2,i}^{\mathrm{norm}} + 0.05)$, normalised to unit mean) that forces the network to prioritise accuracy on resistive targets.

Table S10: DualTCN predictions on seven published MCSEM survey models. MAPE = mean absolute percentage error across all four parameters. DualTCN = unaugmented baseline; DualTCN-W = inverse-σ_2 weighted variant. Forward responses synthesised with `empymod`; single forward pass, no retraining.

Site	Reference	σ_2^* (S/m)	MAPE	MAPE-W	Key improvement
Moderate–high σ_2 ($\geq 0.3\,S/m$)					
GoM Gemini	Constable and Srnka [10]	0.500	10.8%	**6.0%**	σ_2: 12.6→13.2%
North Sea	Constable [9]	0.800	**6.8%**	8.7%	—
West Africa	Key [16]	0.300	13.6%	**7.7%**	σ_2: 36.1→21.3%
Resistive targets ($\sigma_2 < 0.1\,S/m$)					
GoM Resistive	Key [15]	0.050	16.9%	**17.0%**	—
Scarborough Gas	Key [16]	0.020	36.4%	**23.3%**	σ_2: 135.6→69.0%
Norwegian Hydrate	Constable [9]	0.010	57.4%	**15.4%**	σ_2: 208.2→41.9%
Brazil Pre-salt	Constable [9]	0.005	285.6%	**6.8%**	σ_2: 1125→3.4%
Overall			61.1%	**12.1%**	5× reduction

DualTCN-Weighted reduces the overall field-benchmark MAPE from 61.1% to **12.1%** (Table S10). The largest improvements are on the hardest models: Brazil pre-salt (285.6% → 6.8%), Norwegian hydrate (57.4% → 15.4%), and Scarborough gas (36.4% → 23.3%). The cost is a moderate increase in overall test-set loss (0.077 → 0.124), reflecting the trade-off between global accuracy and resistive-target focus.

S14. Pseudo-2D Survey Line Demonstration

To assess DualTCN's applicability beyond isolated 1D soundings, we construct a synthetic 4 km tow line comprising 200 stations over a laterally varying two-layer earth model (Fig. S3). The background seafloor conductivity is $\sigma_2 = 0.50$ S/m, with a 1.6 km-wide resistive body ($\sigma_2 = 0.01$ S/m) embedded between 1.2–2.8 km along the line, simulating a hydrocarbon reservoir. Source depth d_1 varies sinusoidally (60–100 m) to mimic realistic bathymetry, and layer thickness d_2 varies between 15–35 m. At each station, a 1D forward response is computed with empymod and DualTCN-Weighted infers the four parameters independently. A lateral median filter (7 stations, ≈140 m) is applied post-inference, consistent with standard practice in 1D inversion profiling [9].

Table S11 summarises the results. Geometric parameters are recovered accurately: d_1 MAPE = 3.8% ($R^2 = 0.935$) and d_2 MAPE = 16.4% ($R^2 = 0.512$). For the target conductivity σ_2, lateral smoothing reduces the overall MAPE from 39.8% (raw) to 33.1%, and the in-body MAPE from 62.8% to 47.6%. The resistive body is clearly delineated in the recovered 2D section, with lateral boundaries correctly located at ∼1.2 and ∼2.8 km. Errors concentrate at transition zones and within the body interior, where the 50:1 contrast ratio ($\sigma_2 = 0.01$ vs. 0.50 S/m) pushes the limits of the model's dynamic range. Background regions are recovered

Table S11: Pseudo-2D survey results (DualTCN-Weighted, 7-station lateral median filter). In-body refers to the resistive target zone (1.2–2.8 km).

Region	Parameter	MAPE (raw)	MAPE (smooth)	R^2
Overall	σ_1	8.3%	8.1%	—
	σ_2	39.8%	33.1%	0.681
	d_1	3.8%	3.8%	0.935
	d_2	16.8%	16.4%	0.512
In-body	σ_2	62.8%	47.6%	—
Background	σ_2	24.5%	23.4%	—

with 23.4% MAPE for σ_2.

This demonstration confirms that path (i)—pseudo-2D profiling with station-by-station 1D inference and lateral smoothing—produces geologically interpretable sections from the current DualTCN architecture without retraining.

S15. Three-Layer Extension

The two-layer earth model is a deliberate simplification. To demonstrate extensibility, we train a three-layer variant (DualTCN-3Layer) on a 1 000 000-sample dataset with earth model: air | seawater (σ_1) | resistive layer (σ_2, thickness h) | basement (σ_3). This represents a hydrocarbon reservoir or gas-hydrate layer sandwiched between seawater and conductive basement—the standard exploration target for MCSEM.

The physics decoder generalises analytically by summing two sigmoid transitions:

$$\sigma(z) = \sigma_1 + (\sigma_2 - \sigma_1)\,\sigma_{\mathrm{s}}\left(\frac{z - d_{\mathrm{sf}}}{\tau}\right) + (\sigma_3 - \sigma_2)\,\sigma_{\mathrm{s}}\left(\frac{z - d_{\mathrm{sf}} - h}{\tau}\right), \tag{S1}$$

where $d_{\mathrm{sf}} = d_1 + d_2$. The DualTCN encoder is retained unchanged; only the prediction head is widened from 4 to 6 outputs. The resulting model (DualTCN-3Layer, 306 K parameters) is trained with the same protocol as the two-layer variant.

Table S12 reports the results. Three findings stand out. First, the well-constrained parameters σ_1 and d_1 transfer with negligible accuracy loss (+15% and −39% RMSE, respectively—d_1 actually *improves*, likely because the richer subsurface structure provides additional constraints on source depth). Second, d_2 (source-to-layer-top) achieves RMSE = 0.170, essentially identical to the two-layer $d_2 = 0.165$, confirming that the Stage-1 encoder generalises across layer counts. Third, the basement conductivity σ_3 is well resolved ($R^2 \approx 0.88$, RMSE = 0.100), demonstrating that the late-time branch successfully extracts information about the deep layer from the very-late-time diffusion tail.

Table S12: DualTCN-3Layer test-set results (150 000 samples, normalised $[0, 1]$ log-space RMSE). The six parameters are: σ_1 (seawater), σ_2 (resistive layer), σ_3 (basement), d_1 (source depth), d_2 (source-to-layer-top), h (layer thickness). Two-layer DualTCN results shown for comparison where applicable.

Param	3L	2L	Notes
σ_1	0.023	0.020	+15%
σ_2	0.215	0.092*	*half-space
σ_3	**0.100**	—	R^2≈0.88
d_1	**0.019**	0.031	−39%
d_2	0.170	0.165	+3%
h	0.254	—	R^2≈0.23
Loss	0.108	0.077	+40%

The resistive-layer conductivity σ_2 (RMSE = 0.215, $R^2 \approx 0.45$) and layer thickness h (RMSE = 0.254, $R^2 \approx 0.23$) are the hardest parameters, reflecting the well-known σ_2–h trade-off in thin-layer MCSEM inversion: a thin, highly resistive layer produces a similar transient to a thicker, moderately resistive one. This ambiguity is physical, not architectural, and is consistent with the resolution limits of classical MCSEM inversion at 20–200 m offsets. Longer receiver offsets (500–1 000 m) would improve depth resolution and are likely prerequisites for four- or more-layer inversion.

Incorporating VTI anisotropy ($\sigma_h \neq \sigma_v$) requires only a parameter-range extension in `empymod` and a wider prediction head; the main constraint is that the inline HED geometry has limited sensitivity to σ_v [15].

S16. Extended Discussion

This section provides the full discussion subsections that were condensed in the main text.

S16.1 Progressive Architectural Improvement

The three best TCN-based variants form a clear progression (Table 3 in the main text; full ranking in Tables S4–S5): P7 (multi-scale branching) reduces the loss by 9.8% over P5a; P8 (hierarchical conditioning) adds another 4.2%; DualTCN (late-time branch plus auxiliary head) contributes a further 13.6%. Cumulatively, DualTCN is 25.3% better than the baseline, with $\bar{R}^2 = \frac{1}{4}(R^2_{\sigma_1} + R^2_{\sigma_2} + R^2_{d_1} + R^2_{d_2}) = 0.877$.

S16.2 Why the Late-Time Branch Helps

Two mechanisms drive the d_2 improvement. The late-time encoder gives the exponential decay regime its own parameters and receptive field, preventing the weak late-time signal from being overwhelmed by the much stronger early-time response. The auxiliary d_{sf} head provides a gradient signal through a target that the network can estimate more reliably than d_2 alone, preventing the Stage-2 head from effectively

ignoring d_2 in favour of the easier σ_2. This auxiliary loss is used only during training and adds nothing at inference time. Notably, the late-time branch contributes just 28 000 parameters—four dilated blocks on 64 samples—yet provides the largest single accuracy gain in the study.

S16.3 The TCN as the Essential Component

Removing the Transformer and keeping only the TCN (P5a) barely affects accuracy (loss: 0.1020 versus 0.1032) while improving throughput by 2.8×. Removing the TCN and keeping only the Transformer (P5b) more than doubles the loss (0.2248). Replacing both with a flat MLP (P5c) quadruples it (0.4450). The two attention-based encoders (P10a, P10b), despite being parameter-competitive with DualTCN, fare no better than P5b.

The overall ranking is DualTCN > P8 > P7 ≈ P1 > P5a ≈ PCRN ≫ iTransformer ≈ Transformer-only > PatchTST ≫ MLP. The dilated TCN's combination of local temporal processing and an exponentially growing receptive field matches the structure of MCSEM transients, where both fine-grained early-time features and long-range late-time decay carry diagnostic information. Residual networks (ResNet) and encoder-decoder architectures (U-Net), widely used in 2D remote sensing, were not evaluated because the input is a fixed-length 1D time series rather than a 2D spatial field; the dilated causal TCN is the natural 1D analogue of these architectures, providing exponentially growing receptive fields through dilation rather than spatial pooling.

S16.4 Why Attention-Based Encoders Underperform

Both iTransformer and PatchTST produce losses > 2.8× worse than DualTCN despite comparable parameter budgets. Extensive hyperparameter exploration (depth, head count, positional encodings, patch sizes; details in Section S5) did not close the gap. The core issue is inductive bias: the MCSEM transient encodes seawater properties in early-time samples and seafloor properties in the late-time decay. Global self-attention conflates these two physically separate regimes, while the iTransformer discards temporal ordering entirely ($R^2_{d_2} = 0.342$, less than half of DualTCN's). DualTCN's dedicated late-time branch hardcodes this temporal structure, and this specialised inductive bias outperforms flexible attention for signals governed by diffusion physics.

S16.5 Full Comparison with the Literature

DualTCN differs from all published deep-learning CSEM methods in three respects. It operates on time-domain transients rather than frequency-domain data, exploiting the temporal structure through a TCN. Its physics decoder enforces the two-layer profile analytically, producing physically consistent outputs by construction and eliminating the non-physical oscillations that discretised models often require post-processing to remove [33]. Inference is a single feedforward pass under 4 ms—qualitatively different from the iterative loops of physics-informed neural networks [27] or conventional solvers.

Table S13: Methodological comparison with published DL-CSEM approaches. Direct accuracy comparison is not possible due to different datasets and evaluation protocols.

	DualTCN (this work)	**Puzyrev [25]**	**Zhang et al. [33]**
Domain	Time-domain	Frequency-domain	Frequency-domain
Earth model	2-layer (+ 3-layer)	Multi-layer (50 nodes)	3D heterogeneous
Training data	1 M synthetics	10 K synthetics	500 K synthetics
Output	4–6 physical params + profile	Discretised profile	3D resistivity
Architecture	Dual-branch TCN (379 K)	CNN (not reported)	3D CNN (large)
Inference	3.5 ms/sample	∼10 ms/sample	∼10 s/sample
Noise robustness	5 experiments	Not tested	Not tested
Amplitude augm.	Curriculum validated	Not addressed	Not addressed
UQ	4 methods compared	None	None

The parametric output also makes DualTCN a natural component of hybrid workflows. Its estimate of $(\sigma_1, \sigma_2, d_1, d_2)$ can serve as the starting model for a conventional inversion [3], replacing an uninformed guess with a physics-constrained estimate already close to the true model.

Comparison with DL-CSEM methods.. Table S13 places DualTCN alongside published deep-learning marine CSEM methods [25, 33, 32, 18]. Direct numerical comparison is not possible because the studies use different earth models, forward operators, acquisition geometries, and evaluation metrics; the table therefore compares methodological scope rather than accuracy. Note that Zhang et al. [33] solves a fundamentally different (3D volumetric) problem at much larger computational cost; the Puzyrev [25] inference time (∼10 ms) is estimated from the reported architecture size, not directly measured. DualTCN's primary differentiators are: (i) time-domain input with a physics-motivated dual-branch encoder, versus frequency-domain input with generic CNNs; (ii) parametric output (four or six physical scalars) versus discretised profiles or volumes, eliminating non-physical oscillations by construction; (iii) comprehensive noise and amplitude robustness characterisation (five augmentation experiments), which no prior DL-CSEM study has attempted; and (iv) a four-method UQ comparison with calibration analysis.

Positioning against alternative inversion paradigms.. DualTCN's supervised, amortised approach occupies a distinct niche relative to several alternative paradigms. Projected Gauss–Newton (PGN) methods with learned priors [1] retain explicit data-misfit control and can handle 2.5D geometries, but require per-sample iteration (hundreds of forward evaluations), making them $\sim 10^3$–$10^4\times$ slower than DualTCN's single feedforward pass. The warm-start benchmark (Section S9 of this Supplement) demonstrates the natural hybrid: DualTCN provides a physics-constrained initial model that accelerates conventional solvers by an order of magnitude.

Unsupervised differentiable-forward approaches (DIP-style regularisation for MT [29]; differentiable MT inversion [5]) avoid the need for labelled training data, but solve one sample at a time—precluding real-time deployment. DualTCN's million-sample training set amortises the computational investment across all future evaluations.

Ensemble Kalman inversion [14] is a derivative-free alternative well suited to compact parameterisations like ours ($N_{\text{params}} = 4$–6), and naturally produces uncertainty estimates through ensemble spread. However, each ensemble update requires multiple forward evaluations, limiting throughput to $\sim$1–10 samples/s—three to four orders of magnitude slower than DualTCN's 76 K samples/s.

The results in Section S16.4 do not constitute a blanket rejection of attention-based architectures for geophysical time series. Rather, they demonstrate that *generic* global attention is poorly suited to MCSEM transients because early- and late-time regimes are not exchangeable. Physics-aware hybrid attention—e.g., windowed attention restricted to within-regime interactions, or regime-conditioned cross-attention—might retain the representational capacity of Transformers while respecting the temporal hierarchy. Exploring such designs is a promising direction for future work.

Public EM datasets such as OpenEM [23] currently emphasise onshore and airborne regimes; articulating the marine-specific gaps those resources do not cover—time-domain transients, multi-receiver amplitude normalisation, and conductive seawater coupling—motivates the synthetic-data approach adopted here and highlights an opportunity for community benchmark development.

Positioning against ATEM deep-learning work.. The closest methodological predecessors are physics-guided neural networks for airborne time-domain EM (ATEM) inversion [21, 20, 8]. Those studies share DualTCN's key ideas: synthetic training data from a 1D forward solver, a physics-constrained output representation, and auxiliary objectives derived from known physical relationships. DualTCN's late-time branch and auxiliary $\hat{d}_{\text{sf}}$ head are direct analogues of the disentangled multi-window encoders and depth auxiliary losses used in recent ATEM work. The critical differences are physical: marine MCSEM operates at 20–200 m source–receiver offsets in a conducting seawater column (up to $\sigma_1 \approx 3\,\text{S}\,\text{m}^{-1}$), whereas ATEM surveys have kilometre-scale offsets in a resistive air layer. This inverts the dominant noise source (galvanic coupling and motion noise in ATEM versus seawater conductivity and geometric spreading in marine CSEM) and changes the relative importance of early- versus late-time windows for each earth parameter. DualTCN's encoder and loss weighting reflect these marine-specific constraints.

Positioning against probabilistic inversion methods.. Invertible neural networks (INNs) and conditional normalising flows [4, 24] learn the full posterior $p(\boldsymbol{\theta} \mid \mathbf{d})$ rather than a point estimate, providing the richest possible uncertainty quantification. Deep ensembles [17] offer a practical alternative that trains multiple networks independently, capturing epistemic uncertainty through inter-model variance. Both approaches have been applied to seismic and potential-field inversion [22, 7]. Section S6 directly compares MC-Dropout against

three alternative approaches: temperature-scaled MC-Dropout [13], split conformal prediction [2], and a five-member deep ensemble [17]. The comparison shows that (i) MC-Dropout's d_2 under-coverage is correctable post-hoc without retraining, (ii) split conformal prediction provides provably valid intervals by construction, and (iii) the deep ensemble, despite its theoretical appeal, is the least calibrated method—inter-model disagreement across five random seeds substantially underestimates the true prediction error. A conditional RealNVP normalising flow [12, 24] is under development as a more expressive posterior approximation and will be reported separately.

References

[1] Abubakar, A., Habashy, T.M., Li, M., Liu, J., 2012. Inversion algorithms for large-scale geophysical electromagnetic measurements. Inverse Problems 28, 123012. doi:10.1088/0266-5611/28/12/123012.

[2] Angelopoulos, A.N., Bates, S., 2023. Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning 16, 494–591. doi:10.1561/2200000101.

[3] Araya-Polo, M., Jennings, J., Adler, A., Dahlke, T., 2018. Deep-learning tomography. The Leading Edge 37, 58–66. doi:10.1190/tle37010058.1.

[4] Ardizzone, L., Kruse, J., Wirkert, S., Rahner, D., Pellegrini, E.W., Klessen, R.S., Maier-Hein, L., Rother, C., Köthe, U., 2019. Analyzing inverse problems with invertible neural networks, in: Proc. Int. Conf. Learn. Represent. (ICLR). [Online]. Available: https://arxiv.org/abs/1808.04730.

[5] Bai, P., Vignoli, G., Hansen, T.M., 2024. Unsupervised deep learning magnetotelluric inversion. Geophysical Journal International 237, 1–15. doi:10.1093/gji/ggae005.

[6] Bai, S., Kolter, J.Z., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint ArXiv:1803.01271. [Online]. Available: https://arxiv.org/abs/1803.01271.

[7] Bloem, H., Malinverno, A., Brodie, R., 2023. Posterior sampling of geophysical inverse problems: A workflow using conditional normalizing flows. J. Geophys. Res., Solid Earth 128, e2023JB027271. doi:10.1029/2023JB027271.

[8] Colombo, D., Turkoglu, E., Li, W., Sandoval-Curiel, E., Rovetta, D., 2021. Physics-driven deep learning electromagnetic data inversion with applications to subsurface imaging. Geophysics 86, B225–B245. doi:10.1190/geo2020-0759.1.

[9] Constable, S., 2010. Ten years of marine CSEM for hydrocarbon exploration. Geophysics 75, 75A67–75A81. doi:10.1190/1.3483451.

[10] Constable, S., Srnka, L.J., 2007. An introduction to marine controlled-source electromagnetic methods for hydrocarbon exploration. Geophysics 72, WA3–WA12. doi:10.1190/1.2432483.

[11] Constable, S.C., Parker, R.L., Constable, C.G., 1987. Occam's inversion: A practical algorithm for generating smooth models from electromagnetic sounding data. Geophysics 52, 289–300. doi:10.1190/1.1442303.

[12] Dinh, L., Sohl-Dickstein, J., Bengio, S., 2017. Density estimation using real-valued non-volume preserving (Real NVP) transformations, in: Proc. Int. Conf. Learn. Represent. (ICLR). [Online]. Available: https://arxiv.org/abs/1605.08803.

[13] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: Proc. 34th Int. Conf. Mach. Learn. (ICML), pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html.

[14] Iglesias, M.A., Law, K.J.H., Stuart, A.M., 2013. Ensemble Kalman methods for inverse problems. Inverse Problems 29, 045001. doi:10.1088/0266-5611/29/4/045001.

[15] Key, K., 2009. 1D inversion of multicomponent, multifrequency marine CSEM data: Methodology and synthetic studies for resolving thin resistive layers. Geophysics 74, F9–F20. doi:10.1190/1.3058434.

[16] Key, K., 2012. Marine electromagnetic studies of seafloor resources and tectonics. Surveys in Geophysics 33, 135–167. doi:10.1007/s10712-011-9139-x.

[17] Lakshminarayanan, B., Pritzel, A., Blundell, C., 2017. Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 6402–6413. [Online]. Available: https://arxiv.org/abs/1612.01474.

[18] Li, G., et al., 2025. One-dimensional deep learning inversion of marine controlled-source electromagnetic data. Geophysical Prospecting doi:10.1111/1365-2478.13622. early access.

[19] Liu, B., Guo, Q., Li, S., Liu, B., Ren, Y., Pang, Y., Guo, X., Liu, L., Jiang, P., 2020. Deep learning inversion of electrical resistivity data. IEEE Trans. Geosci. Remote Sens. 58, 5715–5728. doi:10.1109/TGRS.2020.2969040.

[20] Liu, W., Wang, H., Xi, Z., Zhou, L., 2022. Physics-driven deep learning inversion with application to transient electromagnetic data. IEEE Trans. Geosci. Remote Sens. 60, 5901212. doi:10.1109/TGRS.2021.3120138.

[21] Moghadas, D., 2020. One-dimensional deep learning inversion of electromagnetic induction data using convolutional neural network. Near Surface Geophysics 18, 11–32. doi:10.1002/nsg.12030.

[22] Mosser, L., Dubrule, O., Blunt, M.J., 2020. Stochastic seismic waveform inversion using generative adversarial networks as a geological prior. Mathematical Geosciences 52, 53–79. doi:10.1007/s11004-019-09832-6.

[23] Open-source Geophysical Electromagnetic Modelling, 2021. OpenEM: Open-source electromagnetic modelling and inversion. https://openem.geoscience.earth. Accessed: 2026-03-01.

[24] Papamakarios, G., Nalisnick, E., Rezende, D.J., Mohamed, S., Lakshminarayanan, B., 2021. Normalizing flows for probabilistic modeling and inference. J. Mach. Learn. Res. 22, 1–64. [Online]. Available: https://jmlr.org/papers/v22/19-1028.html.

[25] Puzyrev, V., 2019. Deep learning electromagnetic inversion with convolutional neural networks. Geophysical Journal International 218, 817–832. doi:10.1093/gji/ggz204.

[26] Puzyrev, V., Swidinsky, A., 2021. Inversion of 1D frequency- and time-domain electromagnetic data with convolutional neural networks. Computers & Geosciences 149, 104681. doi:10.1016/j.cageo.2020.104681.

[27] Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707. doi:10.1016/j.jcp.2018.10.045.

[28] Sami, G.M., 2015. The motion of a horizontal electric dipole in a conducting medium. Int. J. Electron. Commun. Comput. Eng. (IJECCE) 6, 149–154. The geometry comprises a horizontal layer with a moving HED source between two homogeneous half-spaces (seawater above, seafloor below), identical to the shallow-tow configuration used in this study.

[29] Sun, J., Li, Y., 2023. Deep image prior for magnetotelluric inversion. Geophysical Journal International 235, 2231–2247. doi:10.1093/gji/ggad345.

[30] Ulyanov, D., Vedaldi, A., Lempitsky, V., 2018. Deep image prior, in: Proc. IEEE Conf. Computer Vision and Pattern Recognition, pp. 9446–9454. doi:10.1109/CVPR.2018.00984.

[31] Werthmüller, D., 2017. An open-source full 3D electromagnetic modeler for 1D VTI media in Python: empymod. Geophysics 82, WB9–WB19. doi:10.1190/geo2016-0626.1.

[32] Zhang, Z., Hu, Z., 2024. Physics-driven deep-learning for marine CSEM data inversion. J. Appl. Geophys. 229, 105497. doi:10.1016/j.jappgeo.2024.105497.

[33] Zhang, Z., Huang, J., Wan, L., 2024. 3-D CSEM data inversion using deep convolutional neural networks: A feasibility study. Geophysics 89, E55–E68. doi:10.1190/geo2023-0201.1.

[34] Ziolkowski, A., Hobbs, B.A., Wright, D., 2007. Multitransient electromagnetic demonstration survey in France. Geophysics 72, F197–F209. doi:10.1190/1.2735802.

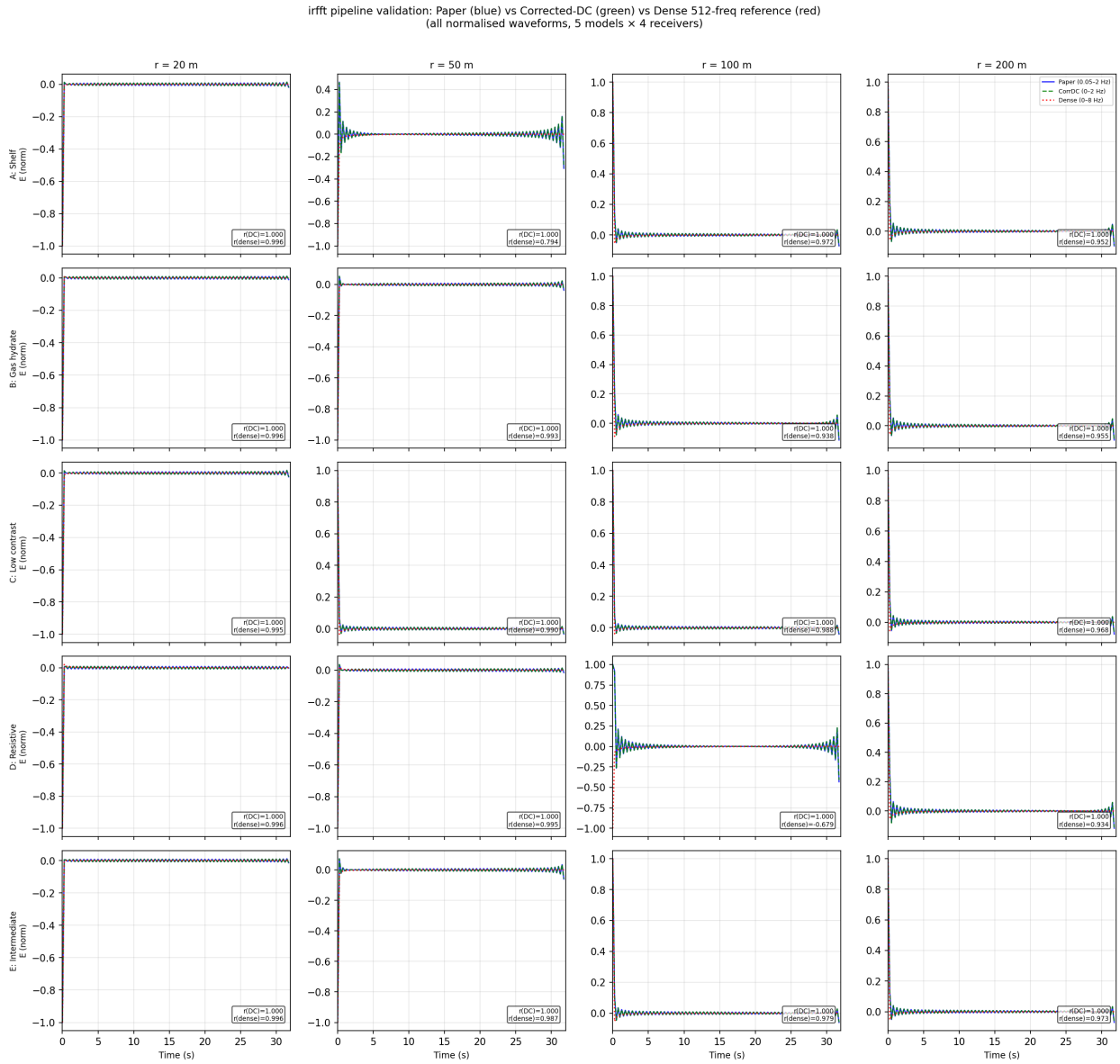

Figure S1: Validation of the irfft synthesis pipeline. Each panel overlays the normalised waveform from the paper's 64-frequency irfft pathway (blue solid), the corrected-DC pathway (green dashed, frequencies 0–2 Hz including a near-DC proxy), and a dense 512-frequency reference (red dotted, 0–8 Hz). Five field-representative models × four receiver offsets = 20 comparisons. The paper and corrected-DC waveforms are virtually identical ($r \geq 0.9998$ in all cases), confirming that the DC misassignment is negligible. Differences with the dense reference (mean $r = 0.97$ excluding one outlier) reflect bandwidth truncation at 2 Hz, not a synthesis artefact.

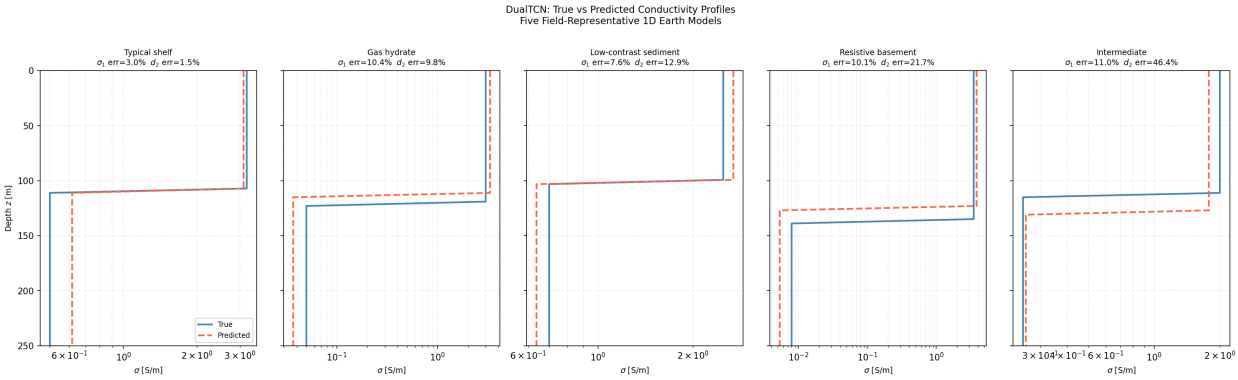

Figure S2: True (blue solid) versus DualTCN-predicted (red dashed) conductivity profiles for the five field-representative 1D models. Each panel reports the relative errors in σ_1 and d_2. The two-step profile shape is recovered in every case. The largest d_2 discrepancy occurs for model E (intermediate contrast, $\sigma_1/\sigma_2 = 8$), consistent with the failure-mode analysis in the main text. Model A (typical shelf) is recovered to within 3% for σ_1 and 1.5% for d_2.

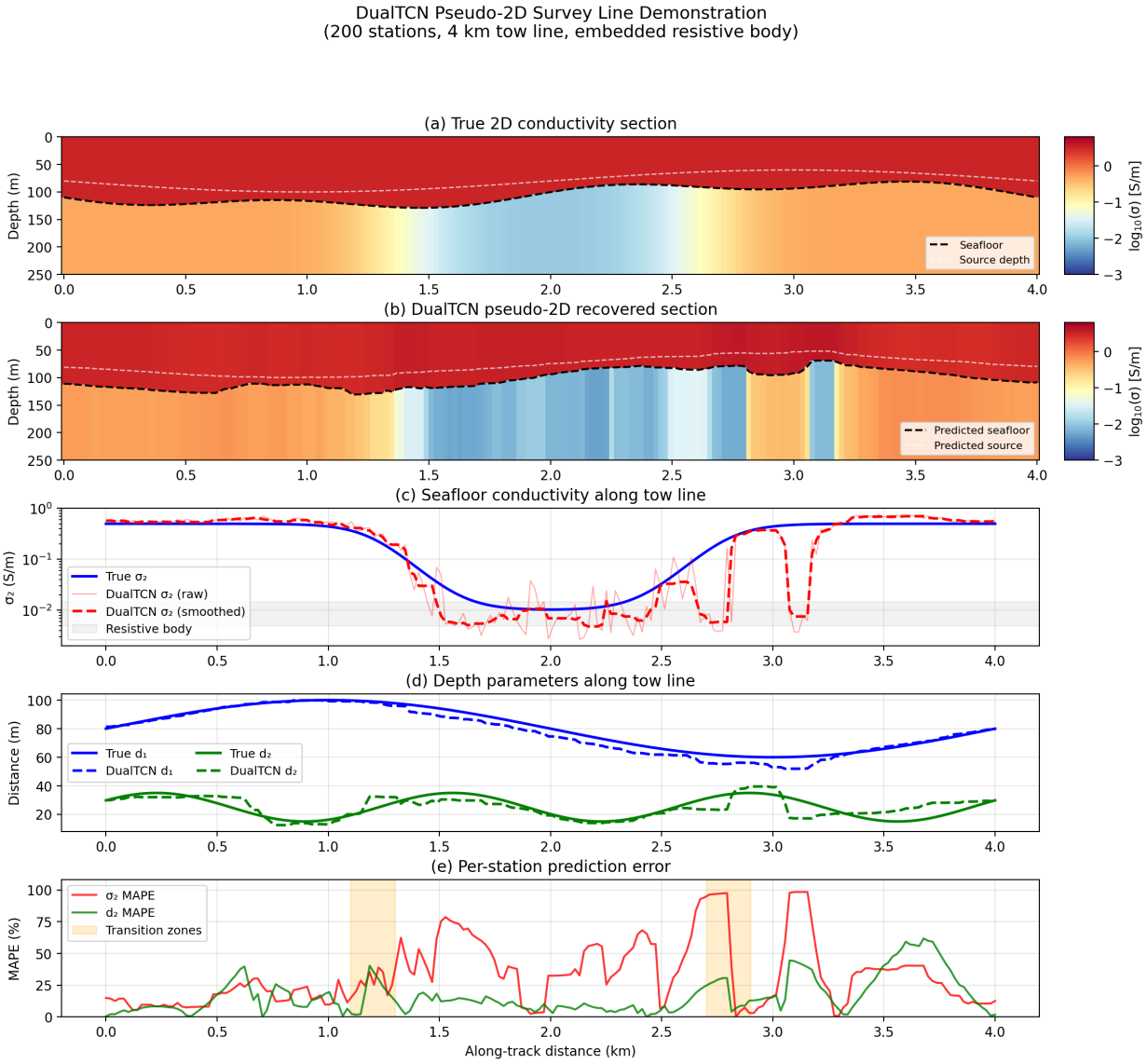

Figure S3: Pseudo-2D survey line demonstration (200 stations, 4 km tow line, embedded resistive body with 50:1 contrast). (a) True 2D conductivity section. (b) DualTCN pseudo-2D recovered section (after lateral median filtering, 7-station window). (c) Seafloor conductivity σ_2 along the tow line: true (blue), raw DualTCN prediction (faint red), and laterally smoothed prediction (dashed red). (d) Depth parameters d_1 and d_2: true (solid) vs. predicted (dashed). (e) Per-station MAPE for σ_2 and d_2; orange bands mark lateral transition zones.